

SEC2_APM1110-FA6-DABU, L.

GitHub: <https://github.com/dabulira20/Prob-ProbDi-FAs/tree/main/FA6>

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I. Geometric Distribution [15 Points]

The geometric distribution models the number of trials required to achieve the first success in a sequence of independent Bernoulli trials, each having a constant probability of success p .

```
# 1. Set the probability of success
p <- 0.2

# 2. Generate 1000 random variables from the geometric distribution.
# Note: R's rgeom() returns the number of FAILURES before the first success,
# so we add 1 to convert it into the number of TRIALS required to achieve
# the first success.
set.seed(123) # for reproducibility
x <- rgeom(1000, prob = p) + 1

# 3. Calculate basic statistics
mean_x <- mean(x)
var_x <- var(x)
sd_x <- sd(x)

# 4. Print the results
cat("Number of trials required to achieve first success:\n")
```

```
## Number of trials required to achieve first success:
```

```
cat("Mean (in 2 decimal places):", round(mean_x, 2), "\n")
```

```
## Mean (in 2 decimal places): 4.9
```

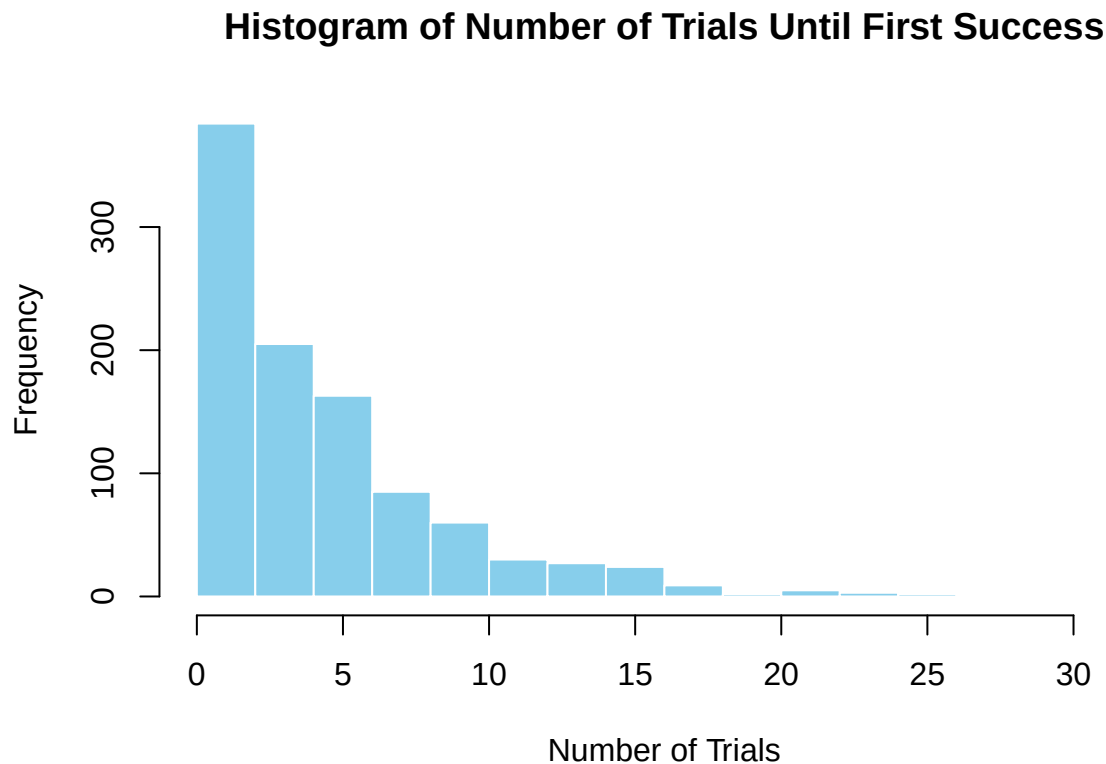
```
cat("Variance (in 2 decimal places):", round(var_x, 2), "\n")
```

```
## Variance (in 2 decimal places): 18.87
```

```
cat("Standard deviation (in 2 decimal places):", round(sd_x, 2), "\n")
```

```
## Standard deviation (in 2 decimal places): 4.34
```

```
# 5. Plot the histogram of the results
hist(x,
     main = "Histogram of Number of Trials Until First Success",
     xlab = "Number of Trials",
     ylab = "Frequency",
     col = "skyblue",
     border = "white")
```



Interpretation: With $p = 0.2$, a success occurs on average every 5 trials (theoretical mean $= 1/p = 5$), which matches closely with the simulated mean above. The histogram is right-skewed, as expected for a geometric distribution — most values cluster at small trial counts with a long tail.

II. Hypergeometric Distribution [15 Points]

A plant manufactures IC chips, 10% of which are defective. Samples are drawn from boxes before shipment. We want the probability that a sample contains **more than 10% defectives**.

The hypergeometric distribution applies here because sampling is done **without replacement** from a finite population (a box of chips).

Parameters used in R's `dhyper(x, m, n, k)`:

- **m** = number of defective chips in the box (successes in population)
- **n** = number of non-defective chips in the box (failures in population)

- k = sample size drawn
- x = number of defectives in the sample

1. Sample of 10 selected from a box of 40 [8 pts]

```
# Box of 40 chips, 10% defective
N1 <- 40 # population size (box size)
defect_rate <- 0.10
m1 <- N1 * defect_rate # number of defective chips = 4
n1 <- N1 - m1 # number of good chips = 36
k1 <- 10 # sample size

# "More than 10% defectives" in a sample of 10 means MORE THAN 1 defective,
# i.e., x >= 2 (since 10% of 10 = 1, "more than" excludes exactly 1).
# P(X > 1) = 1 - P(X = 0) - P(X = 1)
prob_40 <- 1 - phyper(1, m = m1, n = n1, k = k1)

cat("Box size: 40 | Defectives:", m1, "| Good:", n1, "| Sample size:", k1, "\n")
```

```
## Box size: 40 | Defectives: 4 | Good: 36 | Sample size: 10
```

```
cat("P(more than 10% defectives in sample) =", round(prob_40, 4), "\n")
```

```
## P(more than 10% defectives in sample) = 0.2559
```

2. Sample of 10 selected from a box of 5000 [7 pts]

```
# Box of 5000 chips, 10% defective
N2 <- 5000
m2 <- N2 * defect_rate # number of defective chips = 500
n2 <- N2 - m2 # number of good chips = 4500
k2 <- 10 # sample size

# Same logic: more than 1 defective out of 10 sampled
prob_5000 <- 1 - phyper(1, m = m2, n = n2, k = k2)

cat("Box size: 5000 | Defectives:", m2, "| Good:", n2, "| Sample size:", k2, "\n")
```

```
## Box size: 5000 | Defectives: 500 | Good: 4500 | Sample size: 10
```

```
cat("P(more than 10% defectives in sample) =", round(prob_5000, 4), "\n")
```

```
## P(more than 10% defectives in sample) = 0.2639
```

```
cat("Probability (box of 40): ", round(prob_40, 4), "\n")
```

Comparison

```
## Probability (box of 40): 0.2559
```

```
cat("Probability (box of 5000): ", round(prob_5000, 4), "\n")
```

```
## Probability (box of 5000): 0.2639
```

Interpretation: Both probabilities are very close because, as the box (population) size grows much larger relative to the sample size, the hypergeometric distribution converges to the binomial distribution (sampling without replacement behaves almost like sampling with replacement when the population is large). This can be verified by comparing `prob_5000` to `1 - pbinom(1, size = 10, prob = 0.10)`.

```
binom_approx <- 1 - pbinom(1, size = 10, prob = 0.10)
cat("Binomial approximation (large population):", round(binom_approx, 4), "\n")
```

```
## Binomial approximation (large population): 0.2639
```